
HR & PEOPLE ANALYTICS · DATA STORYTELLING
PROJECT

The 100,000-Person Question

What a decade of employee records reveals about pay, performance, attrition — and the HR myths the data dismantles.

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**100,000
Employees**

9 Departments

2014 - 2024

21 Visualizations

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Executive Summary

72%

JOB-TITLE PAY GAP
\$4,545 → \$7,799

0.52%

EDUCATION PAY GAP
PHD VS BACHELOR

~0.2%

GENDER PAY GAP
EQUITY ACHIEVED

10.01%

FLAT ATTRITION
10,010 RESIGNED

3.00 / 2.99

SATISFACTION
STAYED VS RESIGNED

r = 0.51

ONLY REAL SIGNAL
PERFORMANCE ↔ SALARY

Every company believes it knows its people: "top performers stay because they're engaged," "we lose people to remote competitors," "PhDs expect more, so we pay them more," "tenure earns loyalty." These stories feel true — but feelings don't survive contact with 100,000 employee records. This report covers a decade across nine departments. What the data delivered was less a confirmation than a demolition derby for HR folklore.

A Company That Grew Up Overnight

The hiring story starts with a jump. In 2014 the company hired roughly **3,100** people. The very next year that number tripled to **9,900**. That is not growth — that is an organisation assembled at industrial speed, the kind of year where half the company has been there less than twelve months and nobody knows where the coffee mugs are.

From 2015 onward the engine settles into a steady hum: roughly 10,000 hires a year for nearly a decade, peaking at **10,100 in 2022**. Then 2024 happens. Hiring drops to **6,900 — a 31% cut**, the only real shock in the timeline. Whatever happened that year — restructuring, a tighter market, belt-tightening — it left a fingerprint.



Figure 1 — Hiring trend, 2014-2024: steep ramp, long plateau, 2024 contraction.

The people those years brought in are remarkably evenly spread. Marketing is the largest department at **11,216** employees; Engineering the smallest at **10,956** — a difference of **260 people, about 2.3%**. Average tenure sits at **4.48 years**, distributed almost perfectly evenly from years 1 to 10. This is a stable, mature, evenly-built organisation — which is exactly why what comes next is so surprising.

SIGNAL WORTH WATCHING

The 2024 cliff

The hiring dip coincides with the highest remote-work attrition (10.28% at 75% remote). A restructuring signal worth a dedicated post-mortem.

The Compensation Puzzle

Everyone "knows" the hierarchy of pay: education at the top, then experience, then gender gaps, then role. The data kept the scorecard rather differently.

Education: a tie, not a ladder

PhDs average **\$6,431.55**. Bachelors average **\$6,398.42**. That is a **\$33 difference per month — 0.52%** — across something that costs years of life and five or six figures of tuition. The most educated person in the company earns about a dollar a day more than the least educated. Education gets you in the door; it does not set your salary.

Gender: a genuine win

Male (\$6,400.48), Female (\$6,404.99) and Other (\$6,414.77) sit within **\$14 of each other — roughly 0.2%**. Pay equity across genders is, for all practical purposes, solved — rare enough to say out loud and protect in every future audit.

Experience: the landing strip

Year 0 earns **\$6,402**; year 10 earns **\$6,348**. Year 2 — not year 10 — is the high point. The entire eleven-year range is a band about **\$70 wide**. Loyalty is not priced in; worse, the longest-tenured group is the lowest-paid. Imagine staying a decade and discovering new hires match your salary on day one.



Figure 2 — Average salary by years at company: essentially flat, with year 10 the lowest.

Job title: where the numbers wake up

Engineers and Managers average **\$7,799** and **\$7,798**. Consultants \$7,152, Developers \$6,490. Then the cliff: Specialists \$5,855, Analysts \$5,196, Technicians **\$4,545**. The gap between top and bottom role is **\$3,254 a month — 72%**.

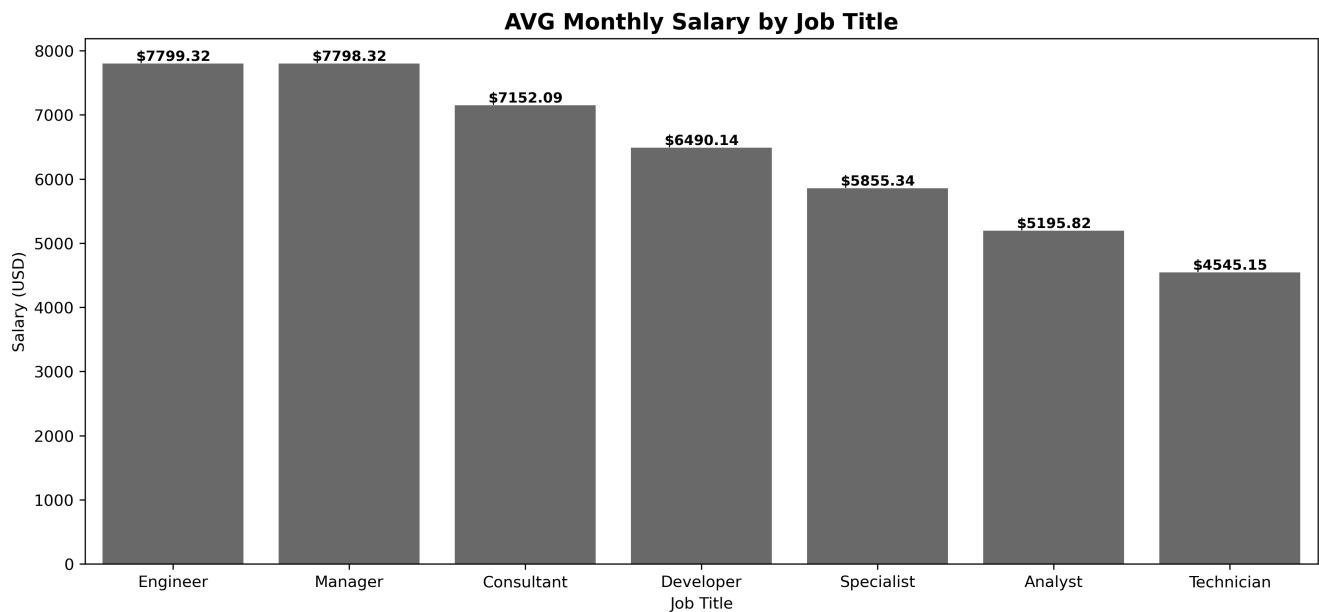


Figure 3 — Average monthly salary by job title: role architecture is the real pay system.

THE PAY SYSTEM

Role, not credentials

The company doesn't pay people for what they know or how long they've stayed — it pays them for what box they sit in. Whether that's fair is a strategy question; whether it's real is not.

The Performance Machine That Produces Noise

Performance scores run on a 1–5 scale. Most companies show a shape — a hump around 3, stars at 5, a small tail of 1s. That shape is how you know managers are actually rating. This company produced a white-picket fence: **20,120 / 20,013 / 19,999 / 19,940 / 19,928** employees at scores 1 through 5 — twenty thousand people in every bucket — a difference of just **192 employees out of 100,000** between "worst" and "best".

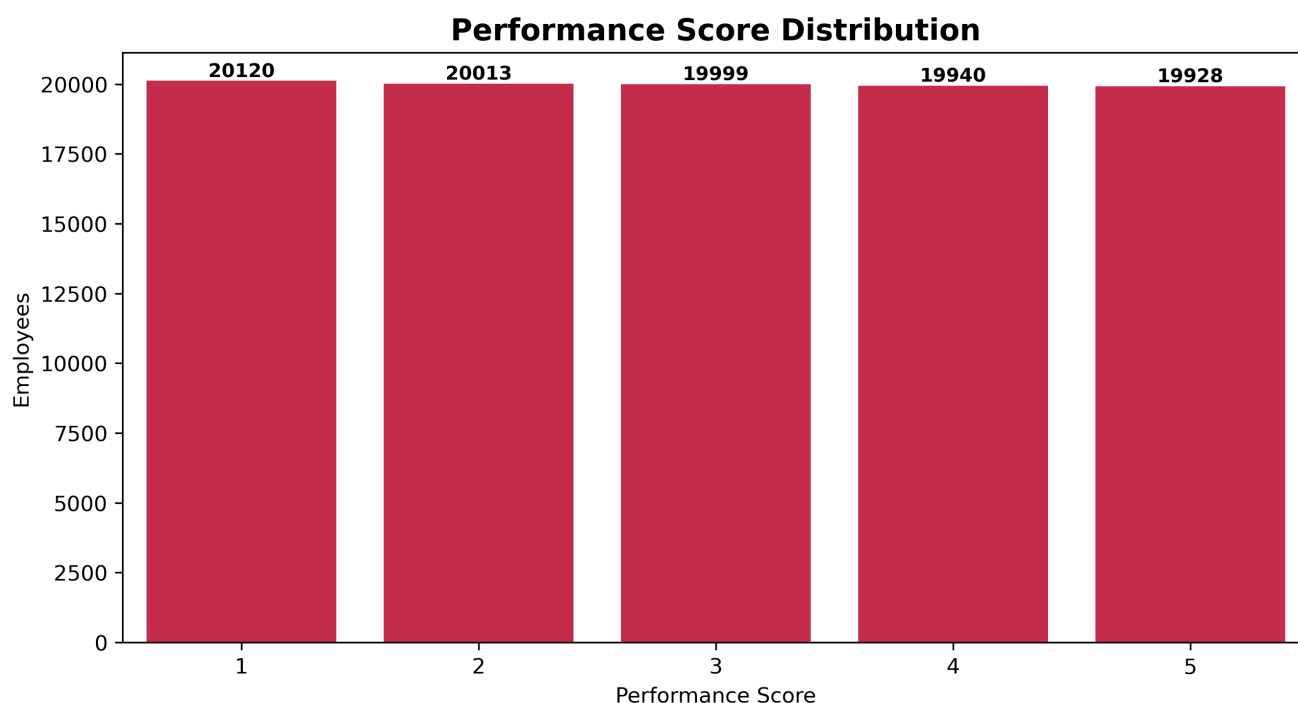


Figure 4 — Performance score distribution: ~20,000 employees in every bucket, 1–5.

Then training. The company invests nearly **50 hours per employee per year** (avg 49.51). Surely high performers get more — or are built by it? Neither. Score 1 averages **49.45** hours; score 5 averages **49.57**. The entire range is **25 minutes of training a year**. Training here is an entitlement distributed evenly, not an intervention aimed at anyone. Promotions don't move ratings either: 3.01 with zero promotions vs 2.99 with one or two.

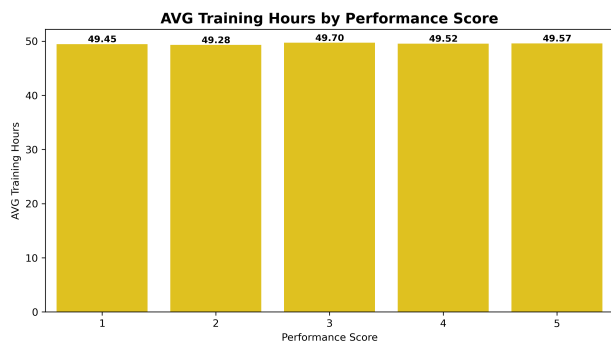


Figure 5 — Training hours flat across all ratings.

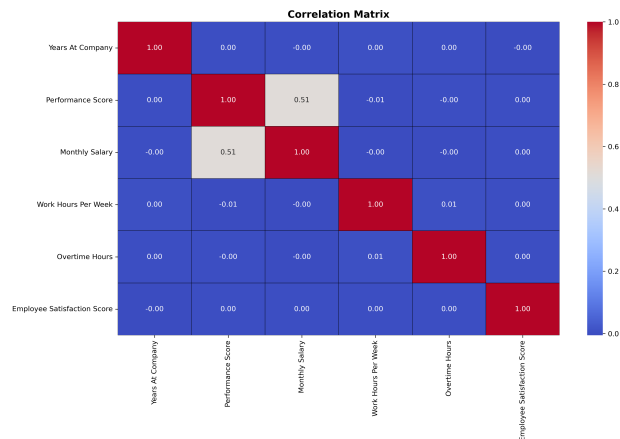


Figure 6 — One correlation survives: performance ↔ salary.

THE STATISTICAL HEART

$r = 0.51$ is the whole game

Across 15 numeric features, Performance ↔ Salary (0.51) is the only relationship that isn't noise. Higher-rated people earn more — so ratings feed pay decisions, yet they measure nothing real, because nothing else in the data responds to them. Everything is three: job titles 2.98–3.01, education 2.99–3.02, gender 2.99–3.00. Reviews need evidence anchors and calibration.

The Attrition Enigma

10,010 people resigned over the decade — an overall attrition rate of **10.01%**. I went looking for who they were. **By department:** Finance highest at **10.54%**, IT lowest at **9.56%** — until you notice the entire spread is **under one percentage point**. **By education:** PhD 10.31% to Master 9.98%. Flat. Turnover is a company-wide constant, not a local-manager problem.

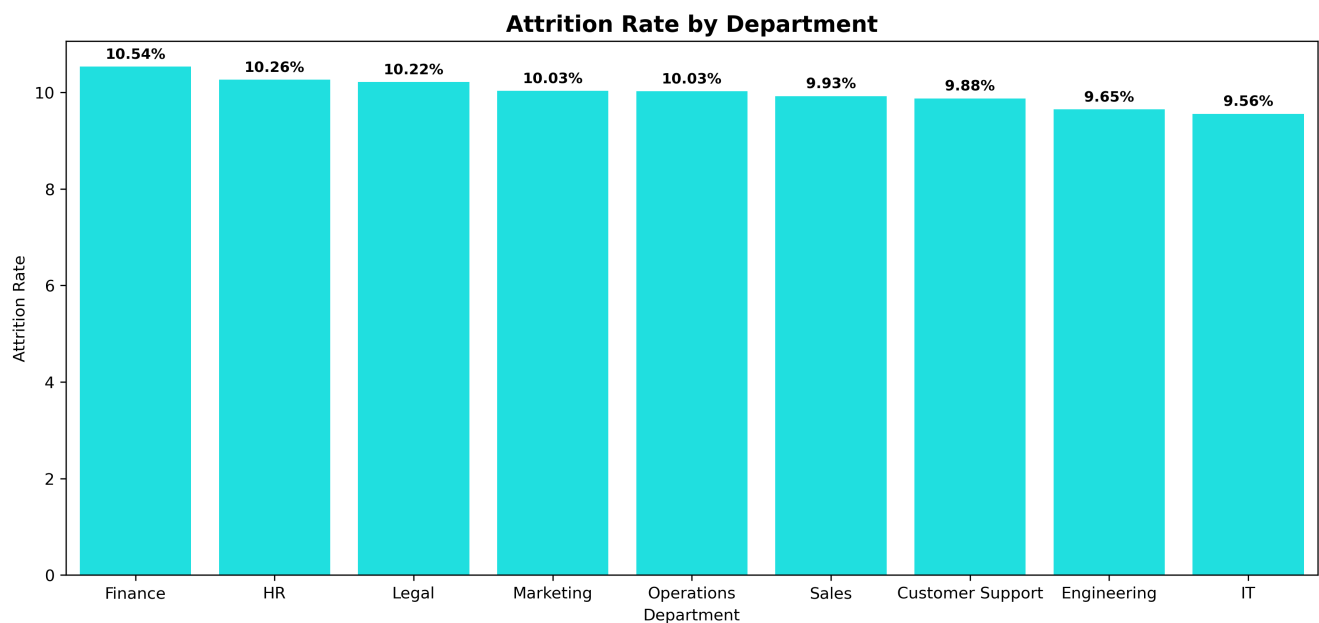


Figure 7 — Attrition by department: under 1 point separates all nine teams.

By remote work: fully in-office (0%) = **9.48%**, 75% remote peaks at **10.28%**, fully remote 10.17%. A story about hybrid isolation writes itself — until you measure the arc: **0.8 percentage points**. Remote policy is a tuning knob, not a driver.

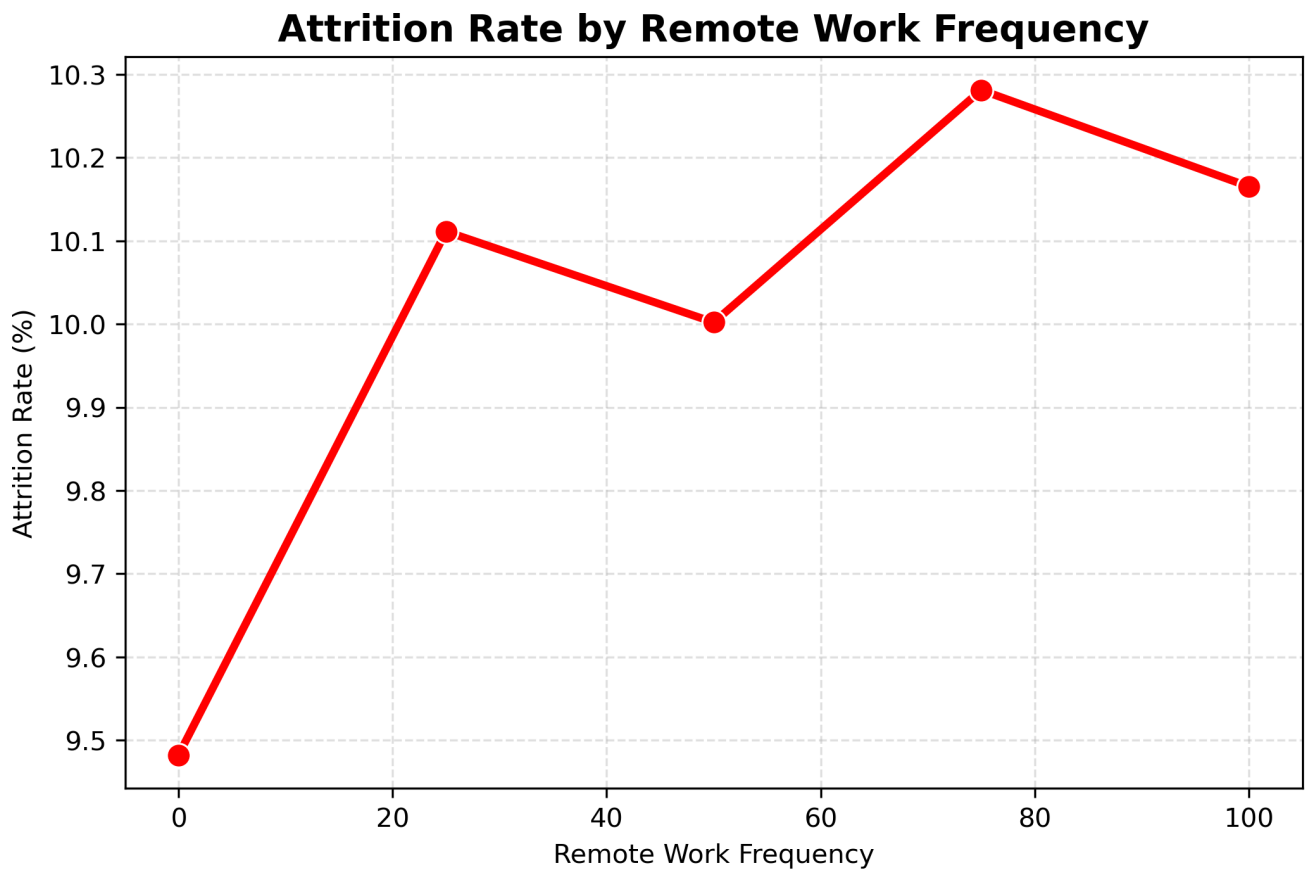


Figure 8 — Attrition vs remote-work frequency: a 0.8-point curve — noise, not cause.

The satisfaction paradox

Then I checked the statistic HR most trusts. Average satisfaction of employees who **stayed**: **3.00**. Of employees who **resigned**: **2.99**. A difference of zero-point-zero-one on a five-point scale. The people who walked out were, statistically, exactly as happy as those who stayed.

AVG Satisfaction: Stayed vs Resigned



Figure 9 — Average satisfaction: stayed 3.00 vs resigned 2.99. The exit signal is missing.

THE GUT-PUNCH

The exit signal is missing

Satisfaction — the metric companies survey, pulse and tie to bonuses — has **zero predictive power** over who leaves. Either it measures mood instead of intent, or people quietly disengage while still ticking "3 - Satisfied." The early-warning system is silent.

Myths vs. Math

HR Belief	What 100,000 Records Show	Verdict	Gap
Higher education → higher pay	PhD \$6,431 vs Bachelor \$6,398	× BUSTED	0.52%
More experience → higher pay	Year 0 \$6,402 · Year 10 \$6,348	× BUSTED	~0%
Training improves performance	49.28–49.70 hrs at every rating	× BUSTED	25 min
Unhappy people quit	Stayed 3.00 vs Resigned 2.99	× BUSTED	0.01
Remote workers leave more	9.48% in-office vs 10.28% at 75%	× MOSTLY	0.8 pt
Ratings differentiate people	~20,000 employees in every score	× BUSTED	uniform
Job title drives pay	\$4,545 Technician → \$7,799 Engineer	✓ CONFIRMED	72%
Pay is gender-equitable	\$6,400–\$6,415 across all groups	✓ CONFIRMED	0.2%

THE BIG IDEA

A dataset where "nothing correlates" is not a failed analysis — it's a diagnosis: people processes are running unmeasured. That's where strategy starts.

What I Would Tell the CEO

1

Rebuild job levelling & publish salary bands

The 72% title gap is the entire pay story. Make promotion paths visible — especially for Technicians (\$4,545) and Analysts (\$5,196), who sit far below the \$6,403 average.

2

Price tenure into compensation

Year-10 employees earning the least is a self-inflicted turnover wound. Time-in-role increments or skill milestones are cheaper than replacing institutional knowledge.

3

Rebuild the attrition early-warning system

Satisfaction can't tell leavers from stayers (3.00 vs 2.99). Add intent-to-stay pulses, manager-quality signals and compa-ratio flags; build a churn model on the 10,010 actual leavers.

4

Target training instead of rationing it equally

49.5 hours for everyone changes nothing. Aim hours at low-rated employees and measure rating lift quarter-over-quarter vs a control group.

5

Calibrate performance reviews

Uniform ratings drive pay ($r = 0.51$) yet say nothing. Introduce evidence anchors, forced-distribution guidance and manager calibration sessions.

6

Investigate the 2024 anomaly

A 31% hiring drop paired with peak hybrid attrition deserves a post-mortem — starting with Finance, the highest-attrition department (10.54%).

AND PROTECT THE WINS

Gender pay equity at ~0.2%, even-handed staffing across departments (2.3% spread), and a stable, mature tenure mix. Those are uncommon.

**The data didn't give this company more questions.
It gave it a to-do list.**

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